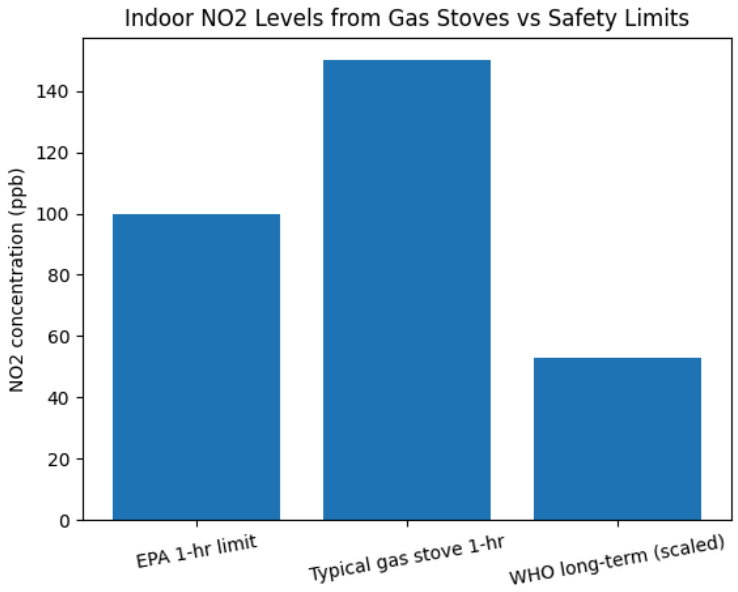
Nitrogen Dioxide: The Problem

Gas and propane stoves are a major but often overlooked source of indoor air pollution. Controlled kitchen experiments and modeling studies have shown that regular stove use can produce 1-hour average nitrogen dioxide (NO_2) concentrations of up to 150 ppb, far exceeding safe levels. For comparison, the U.S. Environmental Protection Agency sets the 1-hour NO_2 safety limit at 100 ppb.



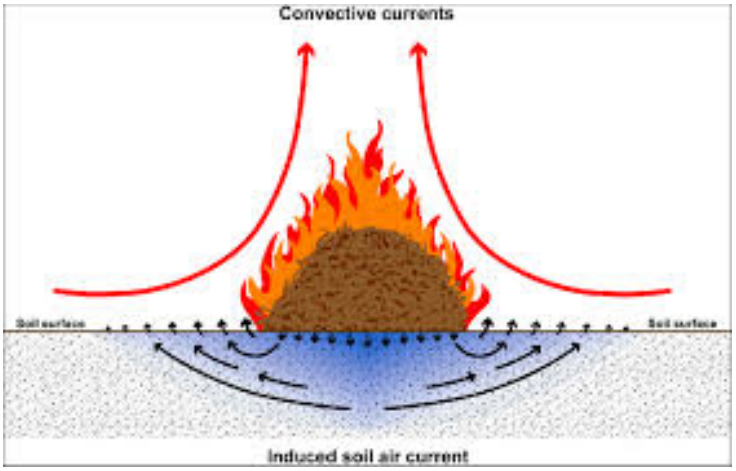
This pollution does not remain confined to the kitchen. Researchers from the Stanford Doerr School of Sustainability found that NO_2 and other combustion pollutants can spread into bedrooms within one hour of stove use and remain at unsafe levels for hours afterward, exposing occupants long after cooking has ended.

The public health consequences are severe. Studies estimate that gas and propane stove emissions contribute to up to 200,000 cases of childhood asthma, with roughly 25% directly caused by NO_2 exposure, according to researchers from the Central California Asthma Collaborative, PSE Healthy Energy, and the Harvard T.H. Chan School of Public Health. Long-term NO_2 exposure from gas stoves is also linked to as many as 19,000 premature deaths per year

in the United States — nearly 40% of the mortality burden of secondhand smoke.

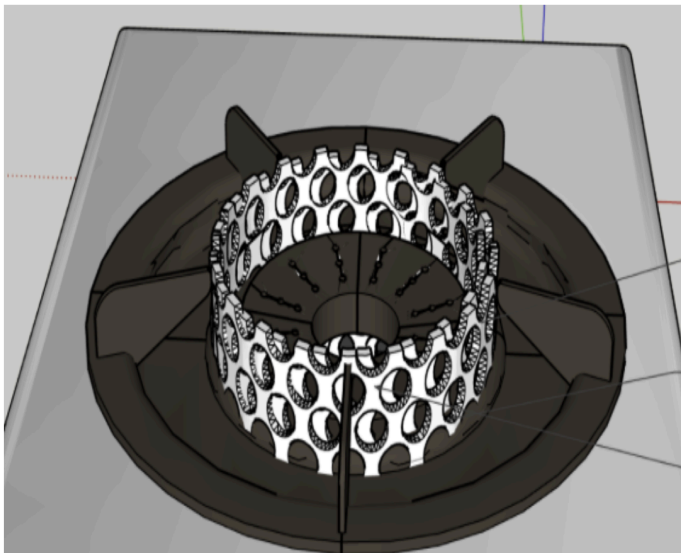
Even small increases in exposure are harmful: gas stove use raises indoor NO_2 levels by about 4 ppb, which is already 75% of the World Health Organization's recommended long-term exposure limit. Short-term cooking peaks frequently exceed health-based standards, placing millions of households at risk.

Cu-Zeolite stove-top filter: Our Solution



When you turn on a gas stove, fuel ignites, creating a flame that reaches very high temperatures (over 1649°C). In these conditions, stable N2 and O2 from the surrounding air combine to form NO2, which is then emitted into the surroundings. The convection currents of air near the stove are displayed in the image below, showing that air is carried in from the sides below the flame and exhaust from the flame, including NO2, is carried up and away, diffusing into the home air and polluting it. Our intended solution is to selectively filter NO2 into 500-1000 times less toxic compounds, such as NO, to improve the safety standards of stove cooking. To achieve this, we propose the use of a Cooper-Zeolite-coated, finely perforated metal structure that will sit as shown on the left, such that the pollutant-filled exhaust current passes through it before exiting into the air. The intent is for the Cu-Zeolite mesh to act as a selective catalytic reducer that allows the selective breakdown of NO2 as described before. Zeolites are crystalline aluminosilicate materials containing nanometer-scale pores

and a negatively charged framework that can hold metal ions, such as, in this case, copper. This capacity of zeolite to host Cu+ ions will allow NO2, which is highly polar and has an unpaired valence electron, to have a strong affinity to the Cu+ while allowing the non-polar stable gases such as O2, CO2, N2, and more to pass uninterrupted, and hence selectively filtering NO2. Furthermore, when the NO2 gains an electron, undergoing reduction and oxidizes Cu(+) to Cu(2+), to become NO2(-). Then, due to the metal body's proximity to the flame and the hot metal grates of the stove that reach high temperatures, the NO2- bonds can be broken and then react with oxygen to form other compounds such as NO and NO3 that can escape the lattice due to their non-polarity and more stable electron configurations. During this process, the Cu(+) that becomes Cu(2+) is also then reduced by radicals such as carbon monoxide and unburned hydrocarbons. These molecules donate electrons to the copper ions, converting Cu^{2+} back into Cu^{+} and restoring the catalyst to its active state. As a result, the copper-zeolite structure operates in a continuous self-regenerating redox cycle, allowing sustained NO_2 conversion without being consumed.



To maximize catalytic efficiency and selectivity, the zeolite support should be of the chabazite (CHA) type, such as SSZ-13 or SAPO-34, which is widely used in modern exhaust after-treatment systems. CHA zeolites have pore sizes (~3.8 Å) that closely match the molecular dimensions of NO_2, enabling strong adsorption while allowing major air components such as N_2 and O_2 to pass through. This confinement enhances the interaction between NO_2 and copper ions while minimizing unwanted side reactions. In addition, CHA frameworks strongly stabilize copper in the Cu^{+} oxidation state and support rapid Cu^{+}/Cu^{2+} redox cycling, which is

essential for sustained NO_2 conversion. The exceptional thermal and hydrothermal stability of CHA zeolites ensures reliable operation under the high-temperature, high-humidity conditions present in stove exhaust, making Cu-CHA the most suitable material for this application.

In this way, the perforated copper-zeolite structure functions as a passive catalytic barrier within the stove's exhaust flow, intercepting and chemically transforming harmful nitrogen dioxide before it disperses into the indoor environment, thereby improving the safety and air quality of gas-stove cooking.